Question	Answer	Marks	Partial Marks
5(a)	3.5, 15, 3.9	**3**	**B1** for each
5(b)	Correct graph	**5**	**B4** for correct curves but branches joined or touching y-axis or **B3FT** 10 or 11 points or **B2FT** for 8 or 9 points or **B1FT** for 6 or 7 points **B1indep** two separate branches not touching or crossing y-axis
5(c)	0.5 to 0.6 and 1.3 to 1.6	**2**	**B1** for each or both correct but in reverse order
5(d)	1	**1**	
5(e)(i)	$y = 3x + 1$ ruled and 0.3 to 0.49	**3**	**B2** for correct ruled line that crosses *their* curve or **B1** for $y = 3x + 1$ soi or freehand line or ruled line with gradient 3 or with y – intercept at 1 (but not $y = 1$)
5(e)(ii)	[a =] –6 [b =] –2 [c =] –4	**3**	**M2** for $x^4 + 2 - 4x = 6x^3 + 2x^2$ or better seen or **B1** for each correct value to a maximum of 2 marks If 0 scored, **SC1** for answer [a =] 6,[b =] 2 and [c =] 4 or for $x^5 + 2x - 4x^2 = 6x^4 + 2x^3$ or better
6(a)(i)	13.9[0…] from cosine rule	**4**	**M2** for $8^2 + 13^2 - 2 \times 8 \times 13\cos 79$ or **M1** for $\cos 79 = \frac{13^2 + 8^2 - BC^2}{2 \times 8 \times 13}$ **A1** for 193 ....
6(a)(ii)	66.6 or 66.60… to 66.65 from sine rule	**3**	**M2** for [$\sin ACB$ =] $\frac{13 \times \sin 79}{their(a)(i)}$ or **M1** for $\frac{\sin ACB}{13} = \frac{\sin 79}{their(a)(i)}$ oe
6(b)(i)	$\frac{1}{2}(x+4)(4x-5)\sin 30 = 70$	**M1**	
	$4x^2 + 16x - 5x - 20 = 280$	**M2**	**Dep on M1** **B1** for $4x^2 + 16x - 5x - 20$ or better
	Leading to $4x^2 + 11x - 300 = 0$	**A1**	with no errors or omissions seen

Question	Answer	Marks	Partial Marks
6(b)(ii)	$\frac{-11 \pm \sqrt{11^2 - 4 \times 4 \times -300}}{2 \times 4}$	**B2**	**B1** for $\sqrt{11^2 - 4(4)(-300)}$ or better or for $\frac{-11+\sqrt{q}}{2\times 4}$ or $\frac{-11-\sqrt{q}}{2\times 4}$
	–10.14 and 7.39	**B2**	**B1** for each or **SC1** for final answers –10.1 or –10.144 to –10.143 **and** 7.4 or 7.393 to 7.394 or –10.14 **and** 7.39 seen in working or for –7.39 **and** 10.14 as final answer
6(b)(iii)	11.4 or 11.39…	**1**	**FT** *their* positive root + 4
7(a)(i)	13	**1**	
7(a)(ii)	3	**2**	**M1** for $\text{h}\left(\frac{10}{30}\right)$ oe soi or $27^{\frac{10}{x}}$
7(a)(iii)	$\frac{7-x}{2}$ oe final answer	**2**	**M1** for $x = 7 - 2y$ or $y - 7 = -2x$ or $7 - y = 2x$ or $-\frac{y}{2} = -\frac{7}{2} + x$ oe
7(b)	0.75 oe final answer	**3**	**M1** for $\frac{10}{2x+1} = 4$ **M1** for $10 = 8x + 4$ or better
7(c)	$\frac{70-19x}{x(7-2x)}$ or $\frac{70-19x}{7x-2x^2}$ final answer	**3**	**M1** for $x + 10(7 - 2x)$ or better isw **B1** for common denominator $x(7 - 2x)$ oe isw
7(d)	3 final answer	**1**	
8(a)(i)	$\frac{m-7}{5}$ oe final answer	**2**	**M1** for $5p = m - 7$ or $p + \frac{7}{5} = \frac{m}{5}$
8(a)(ii)	$[\pm]\sqrt{\frac{y^2-h}{2}}$ or $[\pm]\sqrt{\frac{h-y^2}{-2}}$ oe final answer	**3**	**M1** for first correct step isolate term in p or divide by ±2 **M1** for second correct step FT *their* first step
8(b)(i)	$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$	**1**	
8(b)(ii)	$\begin{pmatrix} -3 \\ -1 \end{pmatrix}$	**1**	